



```
[1]: #Lambda arguments:expression
add=lambda a=3,b=5:a+b
print(add())
```

8

```
[2]: # 15% ,calculating total price after tax has been added.
cal_price = lambda price : price*1.05
print(cal_price(100))
```

105.0

```
[3]: # $1: R16 #Convert us$ to Rand
us_to_zar = lambda us: us*16
print(us_to_zar(10))
```

160

```
[4]: # num1, num2 (multiply two numbers)
Multiply = lambda num1, num2: num1*num2
print(Multiply(10,20))
```

200

```
[5]: Subtract = lambda num_1, num_2: num_1-num_2
print(Subtract(100, 50))
```

50

```
[7]: #Give 10% Discount on a product
final_price = lambda price, discount: price - (price*discount/100)
print(final_price(200,10))
```

180.0

```
: # Lambda function to check even or odd numbers
is_even = lambda x : x%2 == 0
print(is_even(10))
```

True

```
: # Lambda function returns the last digit of a number
last_digit = lambda num: num % 10
print(last_digit(123))
print(last_digit(4567))
```

3

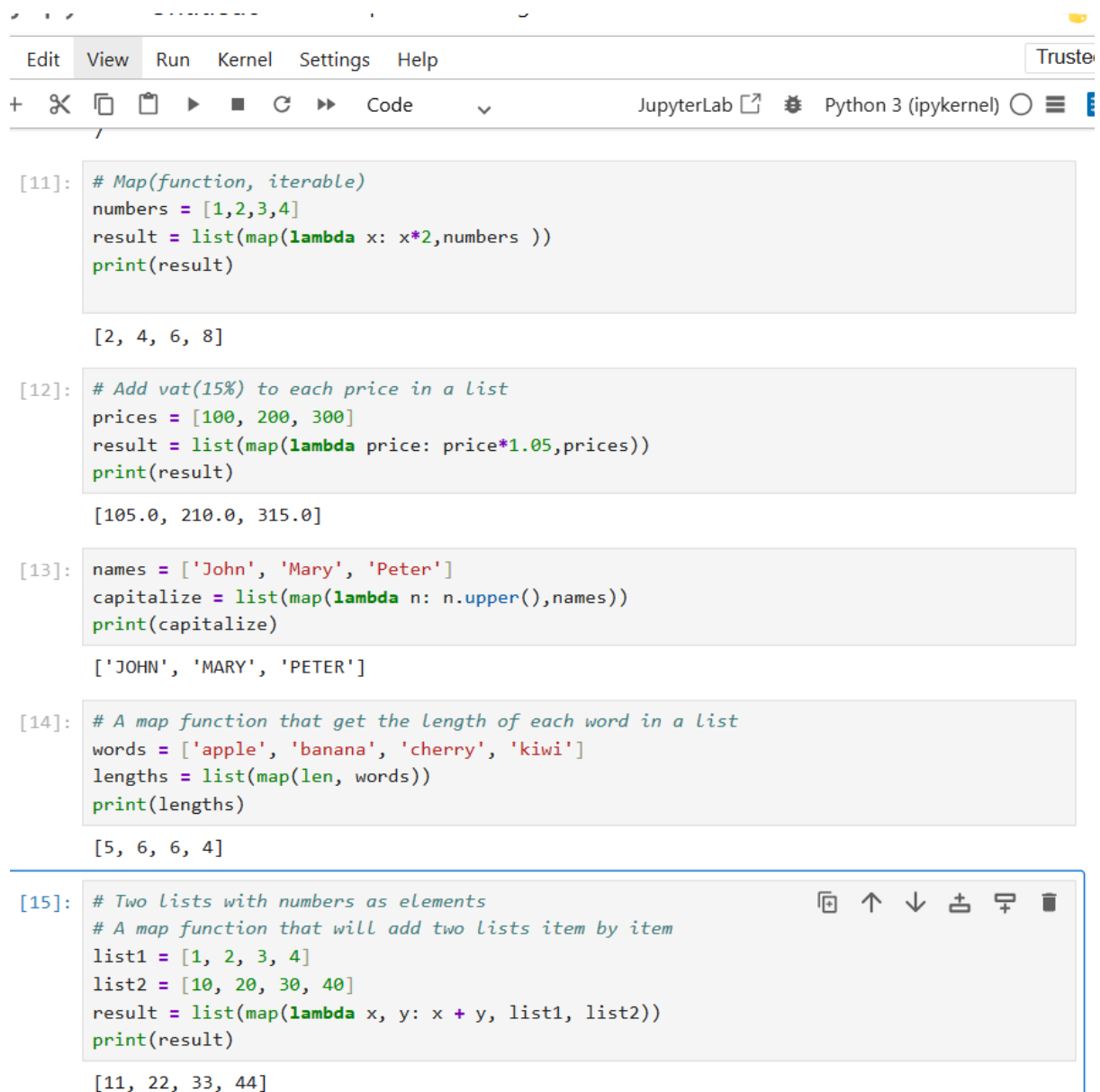
7

```
# Map(function, iterable)
numbers = [1,2,3,4]
result = list(map(lambda x: x*2,numbers ))
print(result)
```

[2, 4, 6, 8]

```
# Add vat(15%) to each price in a list
prices = [100, 200, 300]
result = list(map(lambda price: price*1.05,prices))
print(result)
```

[105.0, 210.0, 315.0]



The screenshot shows a JupyterLab interface with a menu bar (Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and execution. The interface displays five code cells, each with its input code and output result.

[11]: `# Map(function, iterable)`  
`numbers = [1,2,3,4]`  
`result = list(map(lambda x: x*2,numbers ))`  
`print(result)`

[2, 4, 6, 8]

[12]: `# Add vat(15%) to each price in a list`  
`prices = [100, 200, 300]`  
`result = list(map(lambda price: price*1.05,prices))`  
`print(result)`

[105.0, 210.0, 315.0]

[13]: `names = ['John', 'Mary', 'Peter']`  
`capitalize = list(map(lambda n: n.upper(),names))`  
`print(capitalize)`

['JOHN', 'MARY', 'PETER']

[14]: `# A map function that get the length of each word in a list`  
`words = ['apple', 'banana', 'cherry', 'kiwi']`  
`lengths = list(map(len, words))`  
`print(lengths)`

[5, 6, 6, 4]

[15]: `# Two lists with numbers as elements`  
`# A map function that will add two lists item by item`  
`list1 = [1, 2, 3, 4]`  
`list2 = [10, 20, 30, 40]`  
`result = list(map(lambda x, y: x + y, list1, list2))`  
`print(result)`

[11, 22, 33, 44]

```
: #filter(function, iterable)
numbers = [1,2,3,4,5,6,7,8,9,10]
evens = list(filter(lambda x: x%2==0, numbers))
print(evens)
```

```
[2, 4, 6, 8, 10]
```

```
: #filter names longer than 3 letters in a list
names = ['Sam', 'John', 'Amy', 'Peter']
result = list(filter(lambda name: len(name) > 3, names))
print(result)
```

```
['John', 'Peter']
```

```
#filter prices greater than R100
prices = [50, 120, 300, 80, 150]
high_prices = list(filter(lambda p: p>100, prices))
print(high_prices)
```

```
[120, 300, 150]
```

```

]: # modules
import calendar
# Ask for the year from the user
year = int(input('Enter a year(eg 2026):'))
print(calendar.TextCalendar().formatyear(year))

```

Enter a year(eg 2026): 2026

2026

January

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    |    | 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 | 31 |    |

February

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    |    |    |    |    | 1  |
| 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 |    |

March

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    |    |    |    |    | 1  |
| 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 |    |    |    |    |    |

April

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    | 1  | 2  | 3  | 4  | 5  |
| 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 |
| 27 | 28 | 29 | 30 |    |    |    |

May

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    |    |    | 1  | 2  | 3  |
| 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 |

June

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

July

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    | 1  | 2  | 3  | 4  | 5  |
| 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 |
| 27 | 28 | 29 | 30 | 31 |    |    |

August

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    |    |    |    |    | 1  | 2  |
| 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 |    |    |    |    |    |    |

September

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|    | 1  | 2  | 3  | 4  | 5  | 6  |
| 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 28 | 29 | 30 |    |    |    |    |

October

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|

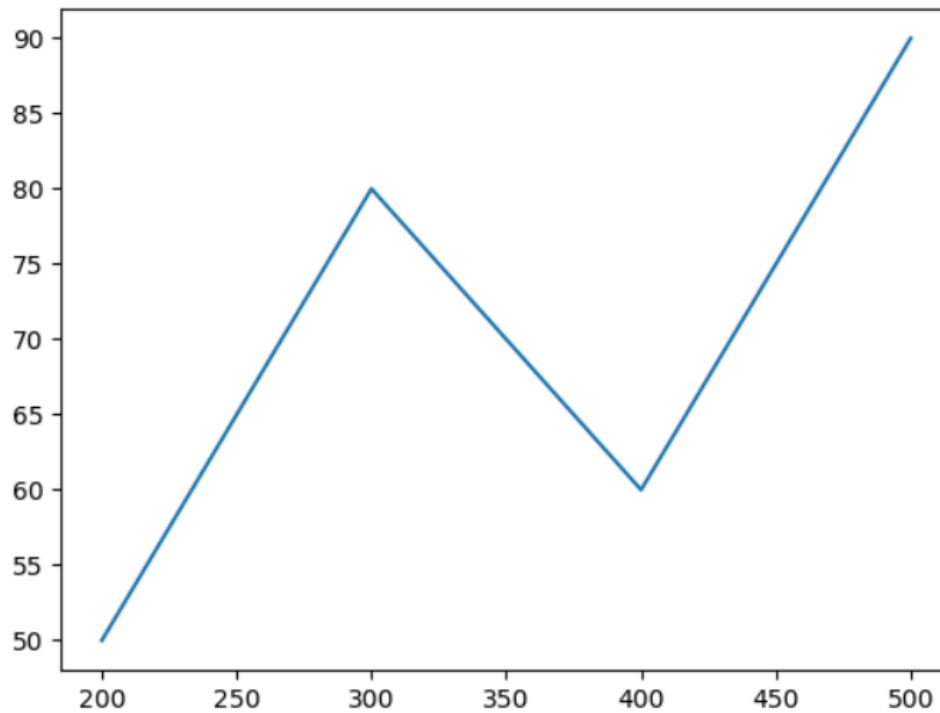
November

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|

December

| Mo | Tu | We | Th | Fr | Sa | Su |
|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|

```
# Data analysis and visualisation
price = [200, 300, 400, 500]
sales = [50, 80, 60, 90]
import pandas as pd
from matplotlib import pyplot as plt
plt.plot(price, sales)
plt.show()
```



```
languages = ['PHP', 'C#', 'Python', 'Java']
students = [100, 120, 150, 200]
plt.bar(languages, students, width = 0.7, colour = 'green')
plt.title('students registered for programming')
plt.xlabel('languages')
plt.ylabel('students')
plt.show()
```

